Chapter 11 Trigonometric Equations - Homework Answers

Chapter revision | Warm-up

1 $\sin x \cos \frac{\pi}{4} + \cos x \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}(\sin x + \cos x)$

2 $\cos 45° \cos 30° - \sin 45° \sin 30° = \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}} = \frac{\sqrt{6} - \sqrt{2}}{4}$

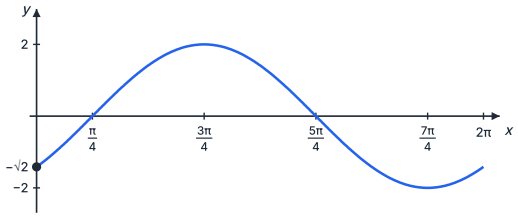
3 $\frac{2\sin^2 \theta}{2 \sin \theta \cos \theta} = \tan \theta$

4 $x = \frac{\pi}{6}$ or $\frac{5\pi}{6}$

5 $0 \leq 2x \leq 4\pi$, so $2x = \frac{\pi}{3}, \frac{4\pi}{3}, \frac{7\pi}{3}, \frac{10\pi}{3}$ and $x = \frac{\pi}{6}, \frac{2\pi}{3}, \frac{7\pi}{6}, \frac{5\pi}{3}$

6 $\sin \theta = \frac{4}{5}, \ \cos \theta = \frac{3}{5}, \ \tan \theta = \frac{4}{3}$

7 Amplitude 2, period 2π. The graph of $y = \sin x$ is stretched vertically by 2 and shifted right $\frac{\pi}{4}$. Intercepts $x = \frac{\pi}{4}, \frac{5\pi}{4}$, y-intercept $-\sqrt{2}$; maximum 2 at $x = \frac{3\pi}{4}$, minimum -2 at $x = \frac{7\pi}{4}$.

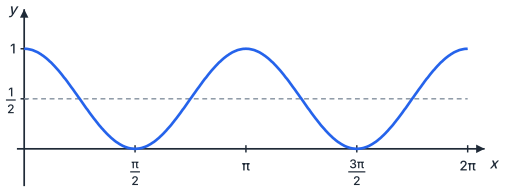


Chapter 11 Trigonometric Equations - Homework Answers

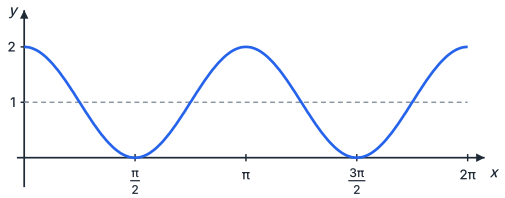
Lesson 11A | Equations involving compound angles

Lesson Examples

C1 **Tutor demo.** $y = \frac{1}{2} + \frac{1}{2}\cos 2x$: period π, amplitude $\frac{1}{2}$, range $0 \leq y \leq 1$.



Your turn. $y = 2\cos^2 x = 1 + \cos 2x$: period π, amplitude 1, range $0 \leq y \leq 2$.



C2 **Tutor demo.** a $4\sin\theta = 3$, so $\theta \approx 48°35'$ or $131°25'$. b $16\sec^2\theta - 16 = 9\sec^2\theta$, $\cos\theta = \pm\frac{\sqrt{7}}{4}$, giving $48°35', 131°25', 228°35', 311°25'$; checking, only $48°35'$ and $131°25'$ satisfy the original equation.
Your turn. $\sin\theta(1 - 2\cos\theta) = 0$, so $\theta = 0°, 60°, 180°, 300°, 360°$.

C3 **Tutor demo 1.** $\cos x(2\sin x + 1) = 0$: $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$.
Your turn 1. $1 - 2\sin^2 x = \sin x$, $(2\sin x - 1)(\sin x + 1) = 0$: $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$.
Tutor demo 2. $\frac{1}{2}\cos x + \frac{\sqrt{3}}{2}\sin x = \sin x$, so $\tan x = \dfrac{1}{2 - \sqrt{3}} = 2 + \sqrt{3}$: $x = \frac{5\pi}{12}, \frac{17\pi}{12}$.
Your turn 2. $\frac{\sqrt{3}}{2}\sin x = \frac{1}{2}\cos x$, $\tan x = \frac{1}{\sqrt{3}}$: $x = \frac{\pi}{6}, \frac{7\pi}{6}$.

C4 **Tutor demo.** $(\tan x - 3)(\tan x + 1) = 0$: $x \approx 1.249, \frac{3\pi}{4}\ (2.356), 4.391, \frac{7\pi}{4}\ (5.498)$.
Your turn. $(\tan x + 2)(\tan x - 1) = 0$: $x = \frac{\pi}{4}\ (0.785), \approx 2.034, \frac{5\pi}{4}\ (3.927), \approx 5.176$.

Textbook Homework

1 b $x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$ or $\frac{3\pi}{2}$

2 b $x = 0, \frac{2\pi}{3}, \frac{4\pi}{3}$ or 2π

3 b $x = \frac{3\pi}{4}$ or $\frac{7\pi}{4}$

4 a $\theta = \frac{\pi}{3}$ or $\frac{4\pi}{3}$ b $\theta = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}, \frac{11\pi}{9}, \frac{13\pi}{9}$ or $\frac{17\pi}{9}$ c $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$ or 2π

5 a $x = \frac{\pi}{2}, \frac{4\pi}{3}, \frac{3\pi}{2}$ or $\frac{5\pi}{3}$ b $x = \frac{\pi}{6}, \frac{\pi}{2}$ or $\frac{5\pi}{6}$
c $x = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}$ or 2π d $x = 0, \frac{\pi}{4}, \frac{3\pi}{4}, \pi, \frac{5\pi}{4}, \frac{7\pi}{4}$ or 2π

6 a $\theta = 90°, 194°29', 270°$ or $345°31'$ b $\theta = 60°$ or $300°$
c $\theta = 41°49', 138°11', 210°$ or $330°$ d $\theta = 106°16'$
e $\theta = 30°, 90°, 150°, 210°, 270°$ or $330°$ f $\theta = 45°, 63°26', 225°$ or $243°26'$

7 b $\theta = \frac{\pi}{12}$ or $\frac{5\pi}{12}$

8 b $x = \frac{\pi}{5}, \frac{3\pi}{5}, \frac{7\pi}{5}$ or $\frac{9\pi}{5}$

9 b $x = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}$ or 2π

10 **a** $\alpha = 60°$ or $300°$ **b** $\alpha = 63°26', 135°, 243°26'$ or $315°$ **c** $\alpha = 30°, 60°, 210°$ or $240°$ **d** $\alpha = 60°$ or $300°$

11 **b** $\theta \approx 160°55'$ or $289°5'$. (Squaring also gives $\sin 2\theta = \frac{1}{2}(1+\sqrt{5}) > 1$, which is impossible, and two angles that fail the check.)

12 **a**
$\cos(2\theta+\theta) = \cos 2\theta\cos\theta - \sin 2\theta\sin\theta = (2\cos^2\theta - 1)\cos\theta - 2\cos\theta(1-\cos^2\theta) = 4\cos^3\theta - 3\cos\theta$
.
b With $x = 2\cos\theta$: $x^3 - 3x - 1 = 2(4\cos^3\theta - 3\cos\theta) - 1 = 2\cos 3\theta - 1 = 0$, so $\cos 3\theta = \frac{1}{2}$, $\theta = 20°, 100°, 140°$. The roots are $2\cos 20°$, $2\cos 100° = -2\sin 10°$ and $2\cos 140° = -2\cos 40°$.

Chapter 11 Trigonometric Equations - Homework Answers

Lesson 11A | Equations involving compound angles

Past-Paper Homework - Worked Solutions

PP 1 **Blacktown Boys' High School, 2022, HSC Trial Examination, Mathematics Extension 1, Q11(d)**

$6\sin x\cos x-\cos x=0$

$\cos x\,(6\sin x-1)=0$

$\cos x=0$: $x=\frac{\pi}{2},\frac{3\pi}{2}$. $\sin x=\frac{1}{6}$: $x=\sin^{-1}\frac{1}{6}\approx 0.17$ or $x=\pi-\sin^{-1}\frac{1}{6}\approx 2.97$.

So $x\approx 0.17,\ 1.57,\ 2.97,\ 4.71$. (Dividing by $\cos x$ would lose $\frac{\pi}{2}$ and $\frac{3\pi}{2}$.)

PP 2 **Sydney Grammar School, 2020, Trial Examination, Form VI Mathematics Extension 1, Q12(a)**

$2\sin x\cos x-\sqrt{2}\sin x=0$

$\sin x\,(2\cos x-\sqrt{2})=0$

$\sin x=0$: $x=0,\pi,2\pi$. $\cos x=\frac{\sqrt{2}}{2}$: $x=\frac{\pi}{4},\frac{7\pi}{4}$.

So $x=0,\ \frac{\pi}{4},\ \pi,\ \frac{7\pi}{4},\ 2\pi$.

PP 3 **Ascham School, 2023, Year 12 Mathematics Extension 1 Trial Examination, Q13(a)**

(i)

$$\tan 2x+\tan x=\frac{\sin 2x}{\cos 2x}+\frac{\sin x}{\cos x}=\frac{\sin 2x\cos x+\cos 2x\sin x}{\cos 2x\cos x}=\frac{\sin(2x+x)}{\cos 2x\cos x}=\frac{\sin 3x}{\cos 2x\cos x}$$

(ii) $\tan 2x+\tan x=0$ requires $\sin 3x=0$ (with $\cos 2x\cos x\neq 0$). For $0\le x\le\frac{\pi}{2}$, $0\le 3x\le\frac{3\pi}{2}$, so $3x=0$ or π.

So $x=0$ or $x=\frac{\pi}{3}$. (At $x=\frac{\pi}{3}$, $\cos\frac{2\pi}{3}\cos\frac{\pi}{3}\neq 0$, so it is valid. $x=\frac{\pi}{2}$ is excluded because $\tan x$ is undefined there.)

PP 4 **James Ruse Agricultural High School, 2020, Year 12 Trial HSC Examination, Mathematics Extension 1, Q13(c)**

(i) Write $\sin 4\theta=\sin(3\theta+\theta)$ and $\sin 2\theta=\sin(3\theta-\theta)$. Adding, the $\cos 3\theta\sin\theta$ terms cancel:

$$\sin 2\theta+\sin 4\theta=2\sin 3\theta\cos\theta.$$

Also $\sin 6\theta=2\sin 3\theta\cos 3\theta$. So

$$\begin{aligned}\text{LHS}&=2\sin 3\theta\,(\cos\theta-\cos 3\theta)\\&=2\sin 3\theta\,\big(\cos(2\theta-\theta)-\cos(2\theta+\theta)\big)\\&=2\sin 3\theta\,(2\sin 2\theta\sin\theta)=4\sin 3\theta\sin 2\theta\sin\theta.\end{aligned}$$

(ii) The equation is equivalent to $4\sin 3\theta\sin 2\theta\sin\theta=0$, for $0\le\theta\le\pi$:

$\sin 3\theta=0$: $\theta=0,\frac{\pi}{3},\frac{2\pi}{3},\pi$; $\sin 2\theta=0$: $\theta=0,\frac{\pi}{2},\pi$; $\sin\theta=0$: $\theta=0,\pi$.

So $\theta=0,\ \frac{\pi}{3},\ \frac{\pi}{2},\ \frac{2\pi}{3},\ \pi$.

PP 5 **The King's School, 2021, Year 12 Alternative Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(c)**

(i)

$$\begin{aligned}\sin(x-y)\sin(x+y) &= (\sin x\cos y-\cos x\sin y)(\sin x\cos y+\cos x\sin y)\\ &= \sin^2 x\cos^2 y-\cos^2 x\sin^2 y\\ &= \sin^2 x(1-\sin^2 y)-(1-\sin^2 x)\sin^2 y\\ &= \sin^2 x-\sin^2 y\end{aligned}$$

(ii) By part (i) with $3x$ and x: $\sin^2 3x-\sin^2 x=\sin 2x\sin 4x$. The equation becomes

$$\sin 2x\sin 4x-\sin 4x=0 \quad\Longrightarrow\quad \sin 4x\,(\sin 2x-1)=0.$$

For $0\le x\le\pi$: $\sin 4x=0$ with $0\le 4x\le 4\pi$ gives $x=0,\ \frac{\pi}{4},\ \frac{\pi}{2},\ \frac{3\pi}{4},\ \pi$; $\sin 2x=1$ gives $x=\frac{\pi}{4}$ (already found).

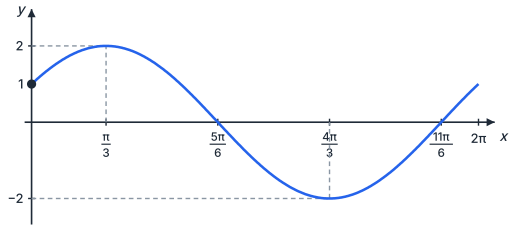
So $x=0,\ \frac{\pi}{4},\ \frac{\pi}{2},\ \frac{3\pi}{4},\ \pi$.

Chapter 11 Trigonometric Equations - Homework Answers

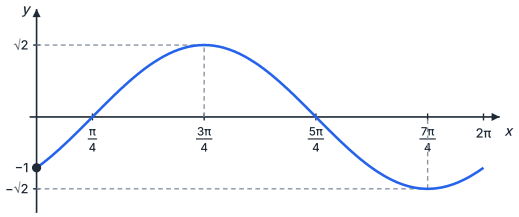
Lesson 11B | The sum of sine and cosine functions

Lesson Examples

C2 **Tutor demo. a** $R\cos\alpha = \sqrt{3},\ R\sin\alpha = 1$, so $R = 2$, $\alpha = \frac{\pi}{6}$: $\sqrt{3}\sin x + \cos x = 2\sin\left(x + \frac{\pi}{6}\right)$. **b** $R\cos\alpha = 1,\ R\sin\alpha = -\sqrt{3}$, so α is in quadrant 4: $2\cos\left(x + \frac{5\pi}{3}\right)$.
Maximum 2 at $x = \frac{\pi}{3}$, minimum -2 at $x = \frac{4\pi}{3}$, x-intercepts $\frac{5\pi}{6}$ and $\frac{11\pi}{6}$, y-intercept 1.

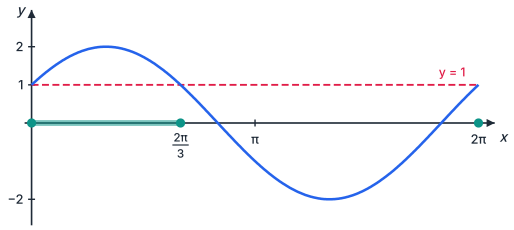


Your turn. $R\cos\alpha = 1,\ R\sin\alpha = 1$: $\sin x - \cos x = \sqrt{2}\sin\left(x - \frac{\pi}{4}\right)$. Maximum $\sqrt{2}$ (at $x = \frac{3\pi}{4}$), minimum $-\sqrt{2}$ (at $x = \frac{7\pi}{4}$); intercepts $\frac{\pi}{4}$, $\frac{5\pi}{4}$, y-intercept -1.



C3 **Tutor demo.** $R\cos\alpha = 12,\ R\sin\alpha = 5$: $R = 13$, $\alpha = \tan^{-1}\frac{5}{12} \approx 22°37'$, so $5\sin x + 12\cos x = 13\cos(x - 22°37')$.
Your turn. $R\cos\alpha = 4,\ R\sin\alpha = 3$: $R = 5$, $\alpha = \tan^{-1}\frac{3}{4} \approx 36°52'$, so $4\sin x + 3\cos x = 5\sin(x + 36°52')$.

C4 **Tutor demo 1.** $\cos(x - \alpha) = \frac{1}{2}$ with $-\alpha \le x - \alpha \le 360° - \alpha$, so $x - \alpha = 60°$ or $300°$: $x \approx 82°37'$ or $322°37'$.
Your turn 1. $\sin(x + \alpha) = \frac{1}{2}$ with $\alpha \le x + \alpha \le 360° + \alpha$, so $x + \alpha = 150°$ or $390°$: $x \approx 113°8'$ or $353°8'$.
Tutor demo 2. $\sin\left(x + \frac{\pi}{6}\right) \ge \frac{1}{2}$ with $\frac{\pi}{6} \le x + \frac{\pi}{6} \le \frac{13\pi}{6}$: $\frac{\pi}{6} \le x + \frac{\pi}{6} \le \frac{5\pi}{6}$ or $x + \frac{\pi}{6} = \frac{13\pi}{6}$. So $0 \le x \le \frac{2\pi}{3}$ or $x = 2\pi$.



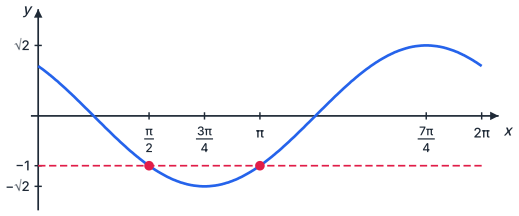
Your turn 2. $\sin\left(x - \frac{\pi}{4}\right) > \frac{1}{\sqrt{2}}$ with $-\frac{\pi}{4} \le x - \frac{\pi}{4} \le \frac{7\pi}{4}$: $\frac{\pi}{4} < x - \frac{\pi}{4} < \frac{3\pi}{4}$, so $\frac{\pi}{2} < x < \pi$.

Textbook Homework

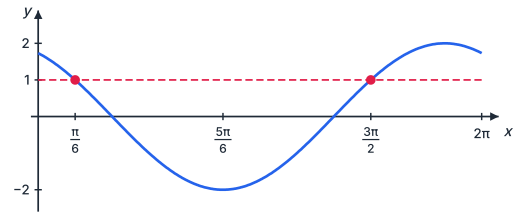
1 **a** $R = 2,\ \alpha = \frac{\pi}{3}$ **b** $R = 3\sqrt{2},\ \alpha = \frac{\pi}{4}$

2 **a** $R = 13,\ \alpha \approx 22°37'$ **b** $R = 2\sqrt{5},\ \alpha \approx 63°26'$

3 **b** $A = \sqrt{2}$ **c** $\alpha = \frac{\pi}{4}$ **d** maximum $\sqrt{2}$ when $x = \frac{7\pi}{4}$; minimum $-\sqrt{2}$ when $x = \frac{3\pi}{4}$ **e** $x = \frac{\pi}{2}$ or π **f** amplitude $\sqrt{2}$, period 2π



4 **b** $B = 2$ **c** $\theta = \frac{\pi}{6}$ **d** greatest 2 when $x = -\frac{\pi}{6}$; least -2 when $x = \frac{5\pi}{6}$ **e** $x = \frac{\pi}{6}$ or $\frac{3\pi}{2}$



5 **c** $x \approx 126°52'$

6 **b** $x = 270°$ or $x \approx 306°52'$

7 **a** $3\sin\left(x + \tan^{-1}\frac{2}{\sqrt{5}}\right)$, that is, $C = 3,\ \alpha \approx 41°49'$ **b** $x = 180°$ or $x \approx 276°23'$

8 **a** $x \approx 77°39'$ or $344°17'$ **b** $x \approx 103°29'$ or $156°8'$ **c** $x \approx 30°41'$ or $297°26'$ **d** $x \approx 112°37'$ or $323°8'$

9 **a** (i) $2\cos\left(x + \frac{11\pi}{6}\right)$ (ii) $x = \frac{\pi}{2}$ or $\frac{11\pi}{6}$ **b** (i) $\sqrt{2}\sin\left(x + \frac{3\pi}{4}\right)$ (ii) $x = 0$ or $\frac{3\pi}{2}$
c (i) $2\sin\left(x + \frac{5\pi}{3}\right)$ (ii) $x = \frac{\pi}{6}$ or $\frac{3\pi}{2}$ **d** (i) $\sqrt{2}\cos\left(x - \frac{5\pi}{4}\right)$ (ii) $x = \pi$ or $\frac{3\pi}{2}$

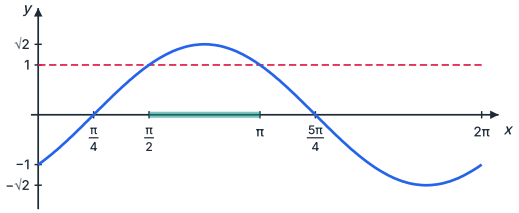
10 **a** (i) $\sqrt{5}\sin(x + 116°34')$ (ii) $x = 270°$ or $x \approx 36°52'$ **b** (i) $5\cos(x - 3.7851)$ (ii) $x \approx 2.63$ or 4.94

11 **a** $x \approx 313°36'$ **b** $x \approx 79°6'$ or $218°59'$ (multiply through by $\cos x$ or $\sin x$ first)

12 **b** $\theta = 0,\ \frac{3\pi}{4},\ \frac{3\pi}{2}$ or $\frac{7\pi}{4}$

13 **a** $x = \frac{7\pi}{12}, \frac{11\pi}{12}$ **b** $x = \frac{\pi}{3}, \frac{4\pi}{3}$ **c** $x = 0, \frac{\pi}{8}, \frac{\pi}{2}, \frac{5\pi}{8}, \pi, \frac{9\pi}{8}, \frac{3\pi}{2}, \frac{13\pi}{8}, 2\pi$

14 **a** (ii) see the sketch; (iii) $\frac{\pi}{2} < x < \pi$



b (i) $\frac{\pi}{2} \le x \le \frac{11\pi}{6}$ (ii) $0 \le x < \frac{\pi}{6}$ or $\frac{3\pi}{2} < x \le 2\pi$ (iii) $\frac{2\pi}{3} < x < \pi$ or $\frac{5\pi}{3} < x < 2\pi$ (iv) $0 \le x \le \frac{\pi}{12}$ or $\frac{17\pi}{12} \le x \le 2\pi$

15 **a** $x-\alpha=\pm\beta+2k\pi$, so $\tan x=\tan(\alpha+\beta)$ or $\tan(\alpha-\beta)$. **b** $R=\sqrt{4+121}=5\sqrt{5}$, $R\cos\alpha=2,\ R\sin\alpha=11$.

c (i) $\cos(x-\alpha)=\frac{2}{\sqrt{5}}=\cos\beta$ where $\tan\beta=\frac{1}{2}$; then $\tan(\alpha-\beta)=\dfrac{\frac{11}{2}-\frac{1}{2}}{1+\frac{11}{4}}=\frac{4}{3}$ and $\tan(\alpha+\beta)=\dfrac{\frac{11}{2}+\frac{1}{2}}{1-\frac{11}{4}}=-\frac{24}{7}$. (iii) $\tan\left(2\tan^{-1}\frac{4}{3}\right)=\dfrac{\frac{8}{3}}{1-\frac{16}{9}}=-\frac{24}{7}$, and $2\tan^{-1}\frac{4}{3}$ is in the second quadrant, so it equals $\pi-\tan^{-1}\frac{24}{7}$.

Chapter 11 Trigonometric Equations - Homework Answers

Lesson 11B | The sum of sine and cosine functions

Past-Paper Homework - Worked Solutions

PP 1 Normanhurst Boys' High School, 2022, Mathematics Extension 1 Assessment Task 4 (Trial Examination), Q9

$R\cos(x-\theta) = R\cos x\cos\theta + R\sin x\sin\theta$. Equating coefficients, $R\cos\theta = 3$ and $R\sin\theta = 2$, so $\tan\theta = \dfrac{R\sin\theta}{R\cos\theta} = \dfrac{2}{3}$.

Answer (C). (A) and (B) have the wrong sign, and (D) is $\dfrac{\cos}{\sin}$ instead of $\dfrac{\sin}{\cos}$.

PP 2 Sydney Grammar School, 2021, Trial Examination, Form VI Mathematics Extension 1, Q12(e)

(i) $R\cos(x-\alpha) = R\cos x\cos\alpha + R\sin x\sin\alpha$, so $R\cos\alpha = 3$, $R\sin\alpha = 2$.

$R = \sqrt{9+4} = \sqrt{13}$, and $\tan\alpha = \frac{2}{3}$ with α acute, so $\alpha \approx 33.69°$:

$3\cos x + 2\sin x = \sqrt{13}\cos(x - 33.69°)$

(ii) $\sqrt{13}\cos(x-\alpha) = \sqrt{13}$, so $\cos(x-\alpha) = 1$, where $-\alpha \le x-\alpha \le 360° - \alpha$. The only solution is $x-\alpha = 0$, so $x = \alpha \approx 33.69°$.

PP 3 Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q12(c)

(i) $A\cos\alpha = 6$, $A\sin\alpha = 8$, so $A = 10$ and $\tan\alpha = \frac{4}{3}$: $\alpha = \tan^{-1}\frac{4}{3} \approx 0.927$.

$6\cos x + 8\sin x = 10\cos\left(x - \tan^{-1}\frac{4}{3}\right)$

(ii) $\cos(x-\alpha) = \frac{1}{2}$, where $-0.927 \le x-\alpha \le 2\pi - 0.927 \approx 5.356$.

$x-\alpha = \frac{\pi}{3}$ or $\frac{5\pi}{3}$ ($-\frac{\pi}{3} \approx -1.047$ is outside the range).

$x = \alpha + \frac{\pi}{3} \approx 1.974$ or $x = \alpha + \frac{5\pi}{3} \approx 6.163$.

PP 4 North Sydney Boys High School, 2023, Year 12 Mathematics Extension 1 Assessment Task 3, Q11(b)

(i) $R\sin(2x+\alpha) = R\sin 2x\cos\alpha + R\cos 2x\sin\alpha$, so $R\cos\alpha = 1$, $R\sin\alpha = \sqrt{3}$. Then $R = 2$ and $\alpha = \frac{\pi}{3}$:

$\sin 2x + \sqrt{3}\cos 2x = 2\sin\left(2x + \frac{\pi}{3}\right)$

(ii) The greatest value is 2, when $\sin\left(2x+\frac{\pi}{3}\right) = 1$. With $\frac{\pi}{3} \le 2x + \frac{\pi}{3} \le 4\pi + \frac{\pi}{3}$: $2x + \frac{\pi}{3} = \frac{\pi}{2}$ or $\frac{5\pi}{2}$, so $x = \frac{\pi}{12}$ or $\frac{13\pi}{12}$.

(iii) $\sin\left(2x+\frac{\pi}{3}\right) = \frac{1}{2}$, with $\frac{\pi}{3} \le 2x + \frac{\pi}{3} \le \frac{13\pi}{3}$:

$2x + \frac{\pi}{3} = \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, \frac{25\pi}{6}$ ($\frac{\pi}{6}$ is too small)

So $x = \frac{\pi}{4}, \frac{11\pi}{12}, \frac{5\pi}{4}, \frac{23\pi}{12}$.

PP 5 Abbotsleigh, 2020, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q13(c)

(i) $R\cos(t+\alpha) = R\cos t\cos\alpha - R\sin t\sin\alpha$, so $R\cos\alpha = 1$, $R\sin\alpha = \sqrt{3}$. Then $R = 2$, $\alpha = \frac{\pi}{3}$:

$\cos t - \sqrt{3}\sin t = 2\cos\left(t + \frac{\pi}{3}\right)$

(ii) Audible when $2\cos\left(t+\frac{\pi}{3}\right) \ge 1$, that is, $\cos u \ge \frac{1}{2}$ where $u = t + \frac{\pi}{3}$ and $\frac{\pi}{3} < u \le \frac{13\pi}{3}$.

$\cos u \ge \frac{1}{2}$ on $\left[-\frac{\pi}{3}, \frac{\pi}{3}\right]$, $\left[\frac{5\pi}{3}, \frac{7\pi}{3}\right]$, $\left[\frac{11\pi}{3}, \frac{13\pi}{3}\right]$. The first interval only touches the range at $u = \frac{\pi}{3}$, which is excluded ($t > 0$).

So $\frac{5\pi}{3} \le u \le \frac{7\pi}{3}$ or $\frac{11\pi}{3} \le u \le \frac{13\pi}{3}$, giving

$\frac{4\pi}{3} \le t \le 2\pi$ or $\frac{10\pi}{3} \le t \le 4\pi$.

Chapter 11 Trigonometric Equations - Homework Answers

Lesson 11C | Using the t-formula to solve equations

Lesson Examples

C1 **Tutor demo.** $10t+2-2t^2=3+3t^2$, so $5t^2-10t+1=0$ and $t=1\pm\frac{2}{5}\sqrt{5}$. With $-90°\le\frac{1}{2}x\le 90°$: $\frac{1}{2}x\approx 62.17°$ or $6.03°$, so $x\approx 124°21'$ or $12°3'$. Check $x=\pm180°$: LHS $=-2\neq 3$.
Your turn. $6t+4-4t^2=1+t^2$, so $5t^2-6t-3=0$ and $t=\frac{1}{5}\left(3\pm 2\sqrt{6}\right)$. $x\approx 115°20'$ or $-41°36'$. Check $x=\pm180°$: LHS $=-4\neq 1$.

C2 **Tutor demo.** $10t-3+3t^2=3+3t^2$, so $t=\frac{3}{5}$ (the t^2 terms cancel): $x\approx 61°56'$. Also $x=180°$: $5\times 0-3\times(-1)=3$. So $x\approx 61°56'$ or $180°$.
Your turn. $4t-1+t^2=1+t^2$, so $t=\frac{1}{2}$: $x\approx 53°8'$. Also $x=180°$: $0-(-1)=1$. So $x\approx 53°8'$ or $180°$.

Textbook Homework

1 **c** $x=0,\ \frac{3\pi}{2},\ 2\pi$

2 **b** $x=0,\ \frac{2\pi}{3},\ 2\pi$

3 **b** $x=90°$ or $x\approx 298°4'$

4 **b** $x=180°$ or $x\approx 67°23'$

5 **b** $t=\frac{1}{2}\left(12\pm\sqrt{144-36}\right)=6\pm 3\sqrt{3}$. **c** $\frac{1}{2}x\approx 84°54'$ or $38°48'$, which double to the given answers.

6 **a** $x=90°$ or $x\approx 12°41'$ **b** $x\approx 36°52'$ or $241°56'$ **c** $x\approx 49°48'$ or $197°35'$ **d** $x=180°$ or $x\approx 280°23'$

7 Multiply by $\cos\theta$: $8\sin\theta-4=\cos\theta$. With $t=\tan\frac{1}{2}\theta$: $16t-4-4t^2=1-t^2$, so $3t^2-16t+5=0$, $(3t-1)(t-5)=0$, $t=\frac{1}{3}$ or 5. Hence $\theta=2\tan^{-1}\frac{1}{3}$ or $2\tan^{-1}5$. ($\theta=\pi$ fails: LHS $=4$.)

8 $x=45°, 225°$ or $x\approx 18.4°, 198.4°$

9 **a** $a(1-t^2)=(1+t)^2$, so $a(1-t)=1+t$ (since $t\neq -1$), giving $t=\dfrac{a-1}{a+1}$. **b** $a=2$: $\tan\frac{1}{2}x=\frac{1}{3}$, $x\approx 36°52'$

10 $6(1-t^2)+34t=18(1+t^2)$ gives $12t^2-17t+6=0$, with roots $t_1=\tan\frac{1}{2}\theta_1$, $t_2=\tan\frac{1}{2}\theta_2$. Then $t_1+t_2=\frac{17}{12}$, $t_1t_2=\frac{1}{2}$, $(t_1-t_2)^2=\frac{289}{144}-2=\frac{1}{144}$, so $t_1-t_2=\frac{1}{12}$ and
$\tan\frac{1}{2}(\theta_1-\theta_2)=\dfrac{t_1-t_2}{1+t_1t_2}=\dfrac{1/12}{3/2}=\frac{1}{18}$.

11 **a** $a(1-t^2)+2bt=c(1+t^2)$ rearranges to $(a+c)t^2-2bt-(a-c)=0$. **b** $\Delta=4b^2+4(a+c)(a-c)=4(a^2+b^2-c^2)\ge 0$. **c** Sum of roots $=\frac{2b}{a+c}$, product $=-\frac{a-c}{a+c}$, so
$\tan\frac{1}{2}(\alpha+\beta)=\dfrac{\frac{2b}{a+c}}{1+\frac{a-c}{a+c}}=\dfrac{b}{a}$.

12 With $t=\tan\frac{1}{2}\theta$ the equation becomes $2kt^2-(k+2)t+1=0$, that is, $(2t-1)(kt-1)=0$. $t=\frac{1}{2}$ gives $\tan\theta=\dfrac{2t}{1-t^2}=\frac{4}{3}$; $t=\frac{1}{k}$ gives $\tan\theta=\dfrac{2k}{k^2-1}$.

Chapter 11 Trigonometric Equations - Homework Answers

Lesson 11C | Using the t-formula to solve equations

Past-Paper Homework - Worked Solutions

PP 1 **Fort Street High School, 2020, Trial Higher School Certificate Examination, Mathematics Extension 1, Q3**

$$\tan\theta\sin\theta = \frac{2t}{1-t^2} \times \frac{2t}{1+t^2} = \frac{4t^2}{(1-t^2)(1+t^2)} = \frac{4t^2}{1-t^4}$$

Answer A. C uses $1+t^4$, which is not $(1-t^2)(1+t^2)$; B and D have the wrong numerator.

PP 2 **Hornsby Girls High School, 2023, Year 12 Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(c)**

$$\frac{2(1-t^2)}{1+t^2} + \frac{6t}{1+t^2} = -2, \quad \theta \neq \pi$$
$$2 - 2t^2 + 6t = -2 - 2t^2$$
$$6t = -4, \qquad t = -\tfrac{2}{3}$$

With $0 \leq \frac{\theta}{2} \leq \pi$: $\tan\frac{\theta}{2} = -\frac{2}{3}$ gives $\frac{\theta}{2} = \pi - \tan^{-1}\frac{2}{3} \approx 2.554$, so $\theta \approx 5.11$.

The t^2 terms cancelled (and the coefficient of $\cos\theta$, 2, is the opposite of the constant -2), so check $\theta = \pi$: $2(-1) + 3(0) = -2$. ✓

So $\theta = \pi$ or $\theta \approx 5.11$.

PP 3 **Kinross Wolaroi School, 2023, Year 12 Mathematics Extension 1 Trial HSC Examination, Q12(b)**

Multiplying by 2: $5\sin x + 2\cos x = 4$. With $t = \tan\frac{x}{2}$:

$$10t + 2(1-t^2) = 4(1+t^2)$$
$$6t^2 - 10t + 2 = 0$$
$$3t^2 - 5t + 1 = 0, \qquad t = \frac{5 \pm \sqrt{13}}{6}$$

With $0° \leq \frac{x}{2} \leq 90°$: $\frac{x}{2} \approx 55.117°$ or $13.084°$, so $x \approx 110°14'$ or $26°10'$.

Check $x = 180°$: $\frac{5}{2}(0) + (-1) = -1 \neq 2$, so it is not a solution.

PP 4 **Normanhurst Boys' High School, 2021, Mathematics Extension 1 (Year 12 Course) Assessment Task 4 (Trial Examination), Q11(c)**

With $t = \tan\theta$: $\sin 2\theta = \frac{2t}{1+t^2}$, $\cos 2\theta = \frac{1-t^2}{1+t^2}$.

$$4t = 2(1+t^2) + 4(1-t^2)$$
$$2t^2 + 4t - 6 = 0$$
$$(t+3)(t-1) = 0$$

$\tan\theta = 1$: $\theta = 45°, 225°$. $\tan\theta = -3$: $\theta \approx 108°26', 288°26'$.

Check $\theta = 90°, 270°$: LHS $= 2\sin 180° = 0$, RHS $= 2 + 4(-1) = -2$, so neither is a solution.

So $\theta = 45°,\ 108°26',\ 225°,\ 288°26'$.

PP 5 **Ascham School, 2023, Year 12 Mathematics Extension 1 Trial Examination, Q14(a)**

$$\frac{6t}{1+t^2} - \frac{5(1-t^2)}{1+t^2} = 5$$

$$6t - 5 + 5t^2 = 5 + 5t^2$$

$$6t = 10, \qquad \tan x = \tfrac{5}{3}$$

$x = \tan^{-1}\tfrac{5}{3} \approx 1.030$ or $x \approx 1.030 + \pi \approx 4.172$.

The t^2 terms cancelled, so check the values the substitution misses, $x = \frac{\pi}{2}$ and $\frac{3\pi}{2}$ (where $2x = \pi, 3\pi$):
$3\sin\pi - 5\cos\pi = 0 + 5 = 5$. ✓ Both are solutions.

So $x = 1.03,\ \frac{\pi}{2},\ 4.17,\ \frac{3\pi}{2}$.

PP 6 **Abbotsleigh, 2020, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q11(a)**

Let $t = \tan 2\theta$, so $\sin 4\theta = \dfrac{2t}{1+t^2}$ and $\cos 4\theta = \dfrac{1-t^2}{1+t^2}$, with $0 \le 2\theta \le 2\pi$.

$$2t + \sqrt{3}(1-t^2) = 0$$

$$\sqrt{3}\,t^2 - 2t - \sqrt{3} = 0$$

$$t = \frac{2 \pm \sqrt{4+12}}{2\sqrt{3}} = \sqrt{3} \text{ or } -\tfrac{1}{\sqrt{3}}$$

$\tan 2\theta = \sqrt{3}$: $2\theta = \frac{\pi}{3}, \frac{4\pi}{3}$, so $\theta = \frac{\pi}{6}, \frac{2\pi}{3}$.

$\tan 2\theta = -\frac{1}{\sqrt{3}}$: $2\theta = \frac{5\pi}{6}, \frac{11\pi}{6}$, so $\theta = \frac{5\pi}{12}, \frac{11\pi}{12}$.

Check $2\theta = \frac{\pi}{2}, \frac{3\pi}{2}$ ($4\theta = \pi, 3\pi$): $0 + \sqrt{3}(-1) \neq 0$, so no extra solutions.

So $\theta = \frac{\pi}{6},\ \frac{5\pi}{12},\ \frac{2\pi}{3},\ \frac{11\pi}{12}$.

PP 7 **Normanhurst Boys' High School, 2023, Year 12 Mathematics Extension 1 Course Assessment Task 4 (Trial Examination), Q12(b)**

With $t = \tan A$: $\cos 2A = \dfrac{1-t^2}{1+t^2}$, $\sin 2A = \dfrac{2t}{1+t^2}$.

$$p(1-t^2) - 2t = 1 + t^2$$

$$(p+1)t^2 + 2t - (p-1) = 0$$

$$\big((p+1)t - (p-1)\big)(t+1) = 0$$

(Check by expanding: $(p+1)t^2 + (p+1)t - (p-1)t - (p-1) = (p+1)t^2 + 2t - (p-1)$.)

So $t = -1$ or $t = \dfrac{p-1}{p+1}$. But $0 \le A \le \frac{\pi}{2}$ means $\tan A \ge 0$, so $t = -1$ is rejected, and

$$t = \frac{p-1}{p+1}.$$

(This needs $p \neq -1$; at $A = \frac{\pi}{2}$, where t is undefined, the equation reads $-p = 1$.)

Chapter 11 Trigonometric Equations - Homework Answers

Chapter 11 Review | Chapter review exercise

1 a $x = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, 2\pi$ b $x = \frac{\pi}{3}, \pi, \frac{5\pi}{3}$ c $x = \frac{7\pi}{6}, \frac{11\pi}{6}$ d $x = \frac{\pi}{3}, \frac{4\pi}{3}$

2 a $\sqrt{2}\sin\left(x - \frac{\pi}{4}\right)$ b $x = \frac{3\pi}{4}$

3 a $2\cos\left(x - \frac{\pi}{6}\right)$ b $x = \frac{5\pi}{6}, \frac{3\pi}{2}$

4 a $3\sin\left(x + \tan^{-1}\frac{\sqrt{5}}{2}\right)$ b $x \approx 41.8°$

5 a $\sqrt{13}\cos\left(x + \tan^{-1}\frac{2}{3}\right)$ b $x \approx 40°12'$ or $252°25'$

6 $x = 0,\ \frac{\pi}{2},\ \frac{3\pi}{2},\ 2\pi$

7 a $14t + 1 - t^2 = 5 + 5t^2$ gives $6t^2 - 14t + 4 = 0$, that is, $3t^2 - 7t + 2 = 0$. b $(3t - 1)(t - 2) = 0$, so $\tan\frac{x}{2} = \frac{1}{3}$ or 2. ($x = \pi$ fails: LHS $= -1$.)

8 $x \approx 1.20$ or 2.87

9 b $x = \frac{\pi}{2},\ \frac{7\pi}{6},\ \frac{3\pi}{2},\ \frac{11\pi}{6}$